



OWNER'S HANDBOOK · COMPLETE EDITION · V2

Converge

FROM INTENT TO EVIDENCE · EVERY PASS · EVERY RELATIONSHIP

Compile intent into
shipped software.

DONE IS PROVEN.

IDENTITY · SETTLEMENT FOLD · AUDIENCE · OWNERS & OPERATORS · SPINE · PASSES 0–8 + OPERATING LAYER · DEPTH · SKILL · CHAT · CLI · GATES

You are the owner. Agents write. Gates prove.

This handbook transfers the full Converge curriculum: every pass, the relationships between **skill · chat interface · cvg CLI**, architecture diagrams, procedures, and owner checklists.

RULE OF THUMB

Four truths

Skill writes · CLI scores · folders remember · **you decide.**

HOW TO READ

Tokens over vibes

A pass is done when the greppable gate token says so – never when work “feels finished.”

STRUCTURE

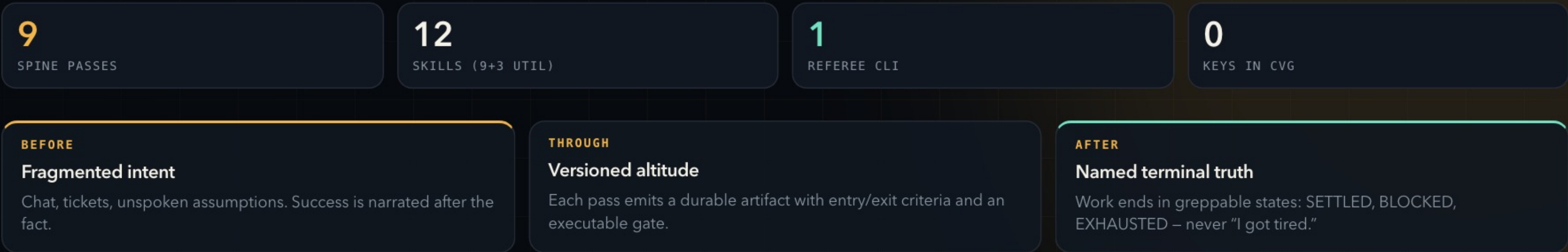
Chapters

Thesis → system → operating layer → each pass vertical → worked example → recovery.

Converged ≠ “feels done.” Converged = gate token green + your required sign-offs.

Autonomy needs a method spine.

Converge compiles intent into software through nine gated passes. Agents build. Machines check. Completion is a state-machine invariant – not an agent’s claim.



Five non-negotiables

- 01

The eval decides done.

A task cannot settle until its evals exit 0. The maximal stub attack (all specs faked) lands all RED.
- 02

The referee is never a player.

`cv``g` holds no API keys and calls no models. Engine CLIs authenticate themselves; the referee only frames, dispatches, and gates.
- 03

Cross-family when it matters.

Pass 4 adversary + Pass 8 tier-2 verify send work to a different model family, prompted to refute – including holdouts the builder never saw.
- 04

A loop with brakes.

Three-axis budgets (iterations · wall-clock · tokens) + stagnation. Only SETTLED, LOCAL_SETTLED, and NO_OP exit zero.
- 05

Evidence over memory.

Position is derived from `cv``g`/`g` folders. Sessions die; disks don't. `cv``g` `next` never trusts chat scrollbar.

Four planes + runtime

Human authority enters; machine evidence returns. Each plane has a different source of truth and failure mode.



Fail-closed hierarchy. Missing authority, failed checks, invalid seals, or unresolved objections block the unattended path. Waivers stay visible and owner-owned.

Three layers. One truth.

One play, three roles. They never swap jobs. Disk evidence survives the session – chat does not.



LAYER	DOES	DOES NOT
Skill	Teaches how to interview, draft, gate	Run itself; hold credentials
Chat agent	Does the work; writes cvg/ ; runs gates	Own business truth; sign for you
cvg CLI	Finds files; runs gate scripts; prints tokens	Write product code; call models
You	Decisions, canonical, barrier, stamps	Memorize every flag (agent can drive)

The cvg/ workspace

Everything Converge “remembers” is folders on disk. **cvg/** is the workspace – not always the git root.



+ .cvg/gate.yaml · brain/ · receipts/ · lessons/

PATH	PASS	CONTENT
.cvg/gate.yaml	init	Write fence a Task-Spec cannot widen
cvg/docs/brd/	0	BRD or no-go
cvg/docs/tech-spec/	1	Falsifiable requirements
cvg/docs/adrs/	2	Terrain ADRs + CONTEXT.md
cvg/swimlanes/	3-4	Plans + objection log
cvg/tasks/	5	Sealed Task-Specs
cvg/execution/	7	Profiles + adapters
cvg/loop/ · receipts/	8	Checkpoints + settlement

Stand up the factory

01 **cvg init**

Creates workspace + `.cvg/gate.yaml`. Idempotent · `CVG_INIT=OK | UNCHANGED`.

02 **cvg setup signing**

Repo-private HMAC key so Task-Specs can be sealed. Never commit the key. Mode 0600 under Git-private state.

03 **cvg setup**

Readiness board + exact next step · `SETUP=READY | INCOMPLETE`.

04 **Optional integrations**

`setup tracker` · `setup key` · `setup repo` · `setup harness` · `doctor`. Secrets stay out of `.cvg/` – only choices are recorded.

Two roots. `CVG_HOME` = installed Converge package. `CVG_PROJECT_ROOT` = consuming project. Workspace may sit below the git root.

Lanes route. They never waive.

`cv`g lane `"..."` picks how much of the spine a change earns. Same final bar everywhere: green eval + path guard.

FAST		NORMAL		FULL	
5 → 7 → 8		1 · 2 · 5 · 7 · 8		0 → 8	
Typo, rename, one-file fix. Small & reversible only.		Most features. Intent + structure without full design theatre.		New idea, new seam, backbone, architecture.	
HARD FLOOR				EFFECT	
auth · money · migrations · secrets · public API				Never FAST	
irreversible / production delete · purge · rotate				Never FAST	
new service · new seam · greenfield · architecture				Forces FULL	
unsigned eval / no sealed spec				No lane dispatches	
>5 files tightens FAST→NORMAL · >15 → FULL pressure				File counts only ever tighten	

cvg next – the descent conductor

Derives where the descent stands from workspace evidence. No state file. No memory. No drift between sessions.

NEXT

Where am I?

Evidence board · NEXT_PASS=N|DONE · skill + pass-prompt.md path · closing gate

PRE N

May pass N start?

Fail-closed door. Prior lane passes must leave artifacts.
PASS_PRE=OK|MISSING

POST N

Did N land?

Folder filled? PASS_POST=OK|INCOMPLETE – then still run the authoritative cvg gate.

Critical rule. **Evidence presence is not a verdict.** A BRD file existing ≠ CHECK_BRD=PASS. Conductor sequences; gates decide. Pre can block forward; it can never grant a pass.

Own the understanding

Optional after any pass. Never blocks the descent. Prevents “the autonomous did it” as an answer to “why is it shaped this way?”

FOUR-PART TREATMENT

Every component

What it is · why shaped · decision encoded · **what breaks downstream** without it.

DURABLE FILE

cvg/docs/lessons/

Survives chat. Onboarding + pre-read for client calls.

GATE

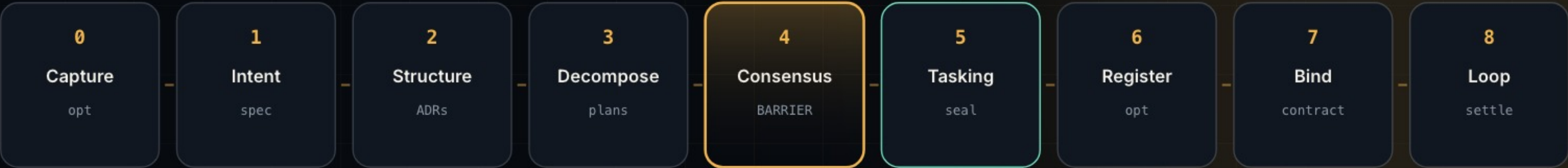
cvg lesson

Seven sections · decisions table · 3-5 quiz Qs · **CHECK_LESSON=PASS**. Optional **--immutable**.

Explain, never reopen. Disagreements become change requests on the owning pass – not silent edits to gated artifacts.

Nine gates. One barrier.

Design half (0-4) makes intent precise. Pass 4 is the last human design sign-off. Build half (5-8) is machine-led proof.



PHASE 1 · DESIGN

Human-led 0-4

BRD → requirements → ADRs → plans → cross-family attack. You grill, approve, sign the barrier.

PHASE 2 · BUILD

Machine-led 5-8

Sealed Task-Specs → optional board → bind → attempt/verify/settle under budgets.

Pass · skill · CLI · token · evidence

#	PASS	SKILL	CLI	TOKEN	EVIDENCE
0	Capture	idea-to-brd	cvg capture	CHECK_BRD	docs/brd/
1	Intent	brd-docs-to-tech-req	cvg intent	CHECK_TECH_SPEC	docs/tech-spec/
2	Structure	tech-req-to-adrs	cvg structure	CHECK_ADR	docs/adrs/
3	Decompose	reqs-to-swimlane-plans	cvg decompose	CHECK_PLAN	swimlanes/
4	Consensus	sketch-plans-adversarial-review	cvg review	CHECK_CONSENSUS	.consensus/
5	Tasking	task-spec	cvg tasks *	TIER=1	tasks/
6	Register	task-specs-to-issues	cvg register	CHECK_REGISTER	tracker
7	Bind	task-to-runtime-contract	cvg bind	CHECK_RUNTIME_CONTRACT	execution/
8	Loop	task-loop	cvg loop	TASK_LOOP	loop/ · receipts/

Capture – raw idea → BRD

Turn a founder thought into a **brief in the owner’s voice** – or a durable no-go. Never the tech-spec. Never architecture.



CONTRACT	
Skill	idea-to-brd
CLI	<code>cvg capture [--draft --no-go]</code>
Token	<code>CHECK_BRD=PASS FAIL DRAFT_OK NOGO_OK</code>
Skip when	Client BRD already exists → enter Pass 1
Altitude	Problem only – no requirements, no stack

Anti-pattern. “Too small for a BRD” is forbidden. Small ideas hide the costliest assumptions. Half-page BRDs still gate.

PASS 0 · EVERY STEP

Scope-check → grill → draft → gate

01 1 · Scope-check

Is this one idea or four platforms? Flag decomposition first. Look up facts in the repo / `cv``g/docs/` – never burn questions on what files know.

02 2 · Grill · frontier rounds

Design tree: pain · do-nothing · goal · scope · success · constraints · pre-mortem. Each round: numbered questions + recommended defaults. One reply answers the whole round (dictation-friendly).

03 Numbers carry provenance

Every number tagged `(measured)` · `(estimated)` · `(guessed)`. A guess spawns an owned open question. Never invent “measured” under pressure.

04 2b · No-go exit

If do-nothing is fine → `cv``g/docs/no-go/<slug>.md` (idea · date · owner · why · reopen condition) · `cv``g capture --no-go`. Cheapest kill in the whole descent.

05 3 · Draft

`cv``g/docs/brd/<slug>.md` from template. Executive summary last (≤60 words). Sign-off present as `_pending_` until you flip to canonical.

06 4 · Gate

`cv``g capture --draft` while writing (never authorizes handoff). You set
canonical + real ISO date
`.cv``g capture → CHECK_BRD=PASS`.

PASS 0 · YOUR ROLE

Owner checklist · Capture

DURING GRILL

- ☐ Answer rounds in one block when possible
- ☐ Prefer honest (guessed) over invented certainty
- ☐ Say “don’t build it” if do-nothing is fine
- ☐ Name one decider if many stakeholders

TO LEAVE PASS 0

- ☐ Sounds like you, not a requirements doc
- ☐ Scope Out is real (not empty / “none”)
- ☐ Every open question has a named owner
- ☐ You set Sign-off to canonical + real date
- ☐ CHECK_BRD=PASS

Forbidden altitude. No “system shall,” no stack decisions, no architecture. Park tech preferences as open questions for Pass 3.

Intent – BRD → testable requirements

Answer: what are we building, and how will we know it works? Never how it is built. Gate is falsifiability, not completeness.



CONTRACT	
CLI	<code>cv9 intent [--draft]</code>
Token	<code>CHECK_TECH_SPEC=PASS ...</code>
Out	<code>cv9/docs/tech-spec/</code>
Boundary	Pass 0 = stakeholder voice · Pass 1 = how-well
Stack	Forbidden – Pass 3 / ADR territory

Ambiguity left here. Is inherited by every later pass. A soft Pass 1 makes ADRs, plans, tasks, and evals soft.

PASS 1 · REWRITE PATTERN

Vague → falsifiable

A requirement needs all three: **measured quantity** · **threshold with unit** · **input condition**. Else → owned open assumption.

VAGUE (FAILS)	FALSIFIABLE (SPIRIT OF THE GATE)
"Dashboard should be fast"	Revenue-by-month ≤ 3s p95 over 12 months of orders
"Data reasonably fresh"	Facts ≤ 15 min lag, ≥ 99% of the day
"Accurate reports"	Reconciled revenue matches ledger within \$0.01 / closed day
"Use DuckDB + dbt"	Park as Pass 3 preference – not a requirement body line

- 01

Ids are forever
R–n requirements · W–n wishes/wonts. Never renumber casually – ADRs and evals cite them.
- 02

Priorities
Not everything is **must**. Use should/could/wont. Deliberate exclusions are honest.
- 03

Blocker gaps
Gate fails closed on missing/blank/sentinel resolutions (TBD, pending, open). Chase the named owner.

Structure – ground the terrain

Lower altitude from *what to build* to what is true about the system we build on.



GROUNDING (KEEP)	DESIGN LEAK (REJECT)
Payments join orders on order_id	Build fct_payments keyed on order_id
Timestamps are TIMESTAMPTZ UTC	Add date_day dimension truncating paid_at
Revenue = paid payments only	Implement revenue_paid as SQL sum

One-line test. Could this sentence have been true before anyone chose a solution? Yes → ADR. No → Pass 3.

ADRs are immutable once accepted

01

Lifecycle

proposed → accepted → deprecated | superseded. Born proposed; accepted at review – never at scaffold time.

02

Supersede, don't rewrite

Reality changes → scaffold-adr.sh --supersede NNNN. Bidirectional links. History stays honest.

03

Worthiness test

Hard to reverse · stood on downstream · could have been otherwise. Else glossary line, not a numbered ADR.

04

CLI

cv g structure · cv g structure --final (nothing left proposed) · CHECK_ADR=OK|FAIL|EMPTY

ACCEPT

☐ Terrain named in 0000-context

☐ Brownfield was run, not summarized

☐ Every ADR cites evidence + re-proof command

☐ Terms in CONTEXT.md · spec_ref: R-n

REJECT

☐ Build/create/implement verbs in ADRs

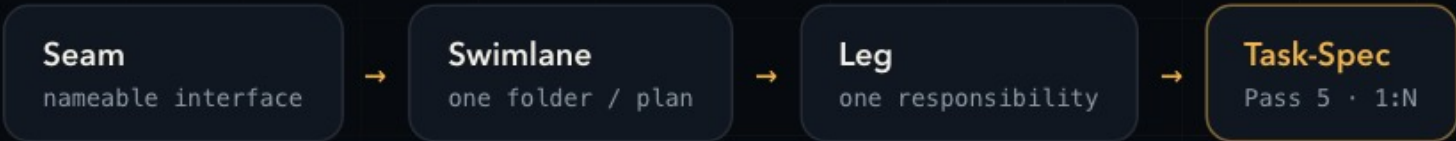
☐ Design leaking into terrain

☐ Closing with records still proposed

☐ Fabricating schema you didn't observe

Decompose – seam → swimlane → leg

Plan altitude only. No tasks. No implementation code. Plans are attacked next – soft spots are expected.



HARDENING	RULE	WHY YOU CARE
H1	Exactly one steel-thread lane	End-to-end path before fattening components
H2	risk= per lane; spike high first	Fatal contracts surface early
H3	owner= stream/team	Conway – wrong owner = coordination tax
H4	Seam evolution stated	Additive vs breaking + coexistence
H5	Cycles broken explicitly	Never a fuzzy two-way seam

Folder shape. cvg/swimlanes/<seam>/_lane.md + leg-NN-<tech>.md. Tech is a swappable label; stable key is swimlane-...-leg-NN. CLI: cvg decompose → CHECK_PLAN=OK.

Consensus – last human design sign-off

A different model family attacks the plans. You FIX or ACCEPT every objection. Only you close the barrier.



COMMAND	ROLE
cvf doctor	≥2 engines · ≥1 cross-family
cvf review --adversary E	Dispatch headless attack
--timeout 900	Raise wall-clock for fat trees
cvf review --check	Provenance re-hash + dispositions

Why here. Signing off before a different model tries to break the plan is signing off on an unexamined mandate. The machine would faithfully build the flaw.

Attack → dispose → sign · when review times out

- 01

ATTACK
Playbook + plans → headless adversary. Default-to-refuted. Emits stamped objection log – never edits plans. Provenance (family + input hashes) stamped by the referee.
- 02

GROUND
Where plans contradict tech-spec or ADRs. Cite both sides. No hand-waving.
- 03

SHARPEN
Every objection: FIX (revise plan in place) or ACCEPT (risk + named owner + reason). Empirical fights → throwaway prototype as evidence.
- 04

BARRIER
Present sharpened plans + dispositions. You sign explicitly. Agent must not start Pass 5 unsigned.
- 05

Timeout ops
Default 300s is often short for 11 files. Use `CVG_TIMEOUT_SECS=900` or `--timeout 900`. Thin tree (steel-thread first). Avoid multi-adversary while debugging latency. Never use same-family Claude-on-Claude as a real barrier.

Tasking – evals first, then seal

Atomic, vendor-neutral, self-verifying units. No eval → not a task yet. After stamp, editing an eval breaks the HMAC seal.

PRE GATE

tasks gate --stamp

Certifies the spec · **VERDICT: DELEGATE** · **TIER=1** · HMAC over eval bodies

POST GATE

tasks accept

Certifies the work after execution · seal recheck · blast radius

FRONTIER

cvg ready

status ready AND all depends_on done – never dispatch past the frontier

EFFORT	ROLE	RULE
XS · S · M · L	Executable leaves	Path budgets ≤1 / ≤2 / ≤3 / ≤5
XL · XXL	Decomposition nodes	Must declare children: – never dispatched
L leaves	execution_backend: glm	Sizing engine absorbs whole range

Author → validate → stamp → lint

- 01

Scaffold

`cv``g` `tasks` `new` `<slug>` `<effort>` – honest sizing. XL/XXL must declare children.
- 02

Evals first

Runnable bash asserting outcomes (state, files, query results) – not “the script ran.” Then goal, touches_paths, budgets, blast radius.
- 03

Holdout

Judge-only criteria under `## Holdout` – excluded from worker brief; reserved for tier-2 verify.
- 04

Validate + stamp

`cv``g` `tasks` `validate` then you authorize `cv``g` `tasks` `gate` `--stamp`.
- 05

Lint backlog

`cv``g` `lint` – cycles, write-surface overlaps. Optional `cv``g` `tasks` `dod` for behavior↔eval matrix.
- Owner stamp policy. You authorize seals. After stamp: never edit evals without re-gate. Silent goalpost moves break every downstream check.

PASS 6 · OPT-IN · TASK-SPECS-TO-ISSUES

Register – specs $\rightleftarrows$ tracker board

One signed spec = one issue. `depends_on` $\rightarrow$ blocked-by. Repo specs remain canonical; the board is a shadow.

01 Only signed_off: true

Un-gated specs never hit the board. Skip and report the rest.

02 Idempotent upsert

Re-runs update; never duplicate. Issue-side marker is the key; `tracker_ref` is a convenience receipt only.

03 Exit Check travels

Issue body carries the close condition. Only a green eval may mark done.

04 Gate

`cvg register --check` $\rightarrow$ 1:1 · DAG · no orphans · no un-gated leaks · `CHECK_REGISTER=OK`

05 Credentials

Referee holds none. `cvg setup key` $\rightarrow$ OS keychain. `.cvg/` stores choices only.

Skip freely. Repo-local queue is valid. Loop reads either board or `cvg/tasks/`.

Bind – freeze the execution contract

One sealed leaf → enforceable profile. Fail closed if the host cannot honor a required control.

7A · CONTRACT

What RUNTIME enforces

`execution-profile.yaml` · path guards · capability envelope · adapter manifests

7B · BRIEF

What MODEL reads

`AGENTS.task.md` – identifiers only, not a novel of context

FENCE

`.cvg/gate.yaml`

Repo write policy a Task-Spec cannot widen. `cvg gate --path`

01 Attest host

`cvg doctor runtime-contract` → OK | DEGRADED | FAIL – know what this machine can enforce

02 Bind

`cvg bind --task <spec>` → CHECK_RUNTIME_CONTRACT=PASS

03 Recheck

`bind --check` is genuinely read-only – recomputes HMAC; never runs task evals

Authority that dies with the task

EPOCH

task-id@spec-sha12

Bound to one signed revision. Epoch change revokes authority.

SCOPE

fs.write = spec paths

Never wider than Task-Spec. Fence cannot be widened by the task.

CLOSURE

settle · block · exhaust

No lingering session permissions after the task ends.

ASSURANCE	MEANING
prevent	Runtime blocks violation (sandbox / pre-tool hook)
detect	Portable postflight catches it after the fact
cannot honor	Gate fails closed unless explicitly waived in the open

Gold / Mint / Coral on the mark. UI state folds: Gold=BOUND · Mint=SETTLED · Coral=RED. Master brand logo stays Forge Gold.

PASS 8 · TASK-LOOP

The Loop – attempt → verify → settle

Drive **one** assigned task to a named terminal state. This loop never picks work – you or CI pass **--issue**.



Four properties of a real loop. Fresh process each attempt · three-axis budgets checked before the call · stagnation beats fixed count · exhaustion is a planned landing with HANDOFF.md.

PASS 8 · TASK_LOOP=

An error is never a success

STATE	EXIT 0?	MEANING · YOUR ACTION
SETTLED	yes	Green + PR – review & merge
LOCAL_SETTLED	yes	Green local commit (external_writes deny) – correct, not failure
NO_OP	yes	Already green on arrival
BLOCKED	no	Human / missing upstream – read blocked report
STALLED	no	Same failure N times – stop burning money; fix cause
EXHAUSTED	no	Budget hit · HANDOFF.md · --resume
CANCELLED	no	cvg/loop/<id>/STOP honored
ERROR	no	Loop unsafe – investigate environment

Scoped, ordered, policy-governed

- 01

Stage only authorized paths

From capability envelope `fs.write` – never `git add -A` (untracked files would slip past diff guards).
- 02

External writes are separate

Default deny → `LOCAL_SETTLED`. Push/PR require policy or `--allow-external-writes`. Commit, push, tracker, PR are four effects.
- 03

Receipt last

A pass receipt never attests a settlement that didn't happen.
- 04

Optional tier-2

`cvg verify --task ... --judge <other-family> · holdout worker never saw · CHECK_VERIFY=UPHELD|REFUTED|UNAVAILABLE`
- 05

Accept + next

`cvg tasks accept · then cvg ready` for next unblocked leaf

Storm-alerts feed · end to end

One concrete product idea through every pass – files, tokens, and your moves.

- 01

Day 0 · Setup

cv^g init · setup signing · setup → READY.cv^g lane "customer storm-alerts feed" → FULL.cv^g next --lane FULL → NEXT_PASS=0.
- 02

Design 0-4

BRD → tech-spec → ADRs → swimlanes (ingest · deliver · steel-thread) → Codex attack → triage objections → you sign barrier
- 03

Build 5-8

Seal steel-thread task first (evals first) · optional Linear register · bind · loop until TASK_LOOP=SETTLED.
- 04

Between tasks

cv^g ready → next unblocked leaf → bind → loop until frontier empty.

What green looks like

PASS	EVIDENCE ON DISK	TOKEN
0	cvg/docs/brd/storm-alerts-feed.md	CHECK_BRD=PASS
1	cvg/docs/tech-spec/...	CHECK_TECH_SPEC=PASS
2	cvg/docs/adrs/* + CONTEXT.md	CHECK_ADR=OK
3-4	swimlanes/** + .consensus/	CHECK_PLAN + CHECK_CONSENSUS
5	cvg/tasks/T-*.md sealed	TIER=1
7	cvg/execution/<id>/	CHECK_RUNTIME_CONTRACT=PASS
8	PR + receipts/	TASK_LOOP=SETTLED

Steering phrases. "FULL lane. cvg next." · "I'll sign when draft is green." · "No build verbs in ADRs." · "Dispatch Codex; I sign." · "Evals first; stamp when I say." · "Report the token only."

Token glossary

TOKEN	MEANS
CHECK_BRD=PASS	Brief handoff-ready
CHECK_TECH_SPEC=PASS	Spec handoff-ready
CHECK_ADR=OK	Terrain recorded
CHECK_PLAN=OK	Swimlane tree sound (machine subset)
CHECK_CONSENSUS=OK	Barrier machine-half green
TIER=1 / DELEGATE	Spec sealed
CHECK_REGISTER=OK	Board mirrors backlog
CHECK_RUNTIME_CONTRACT=PASS	Bound to host
TASK_LOOP=SETTLED	Shipped through outward legs
NEXT_PASS=DONE	Descent complete for that lane
CHECK_LESSON=PASS	Lesson structurally teachable
CHECK_VERIFY=UPHELD REFUTED	Tier-2 independent judge

What only you must do

PASS	YOU MUST PERSONALLY...
0	Answer grill · canonical on BRD · or no-go
1	Close blockers · canonical on tech-spec
2	Accept terrain facts · refuse design-as-ADR
3	Approve seams / steel thread / owners
4	Resolve objections · sign the barrier
5	Authorize stamp · place holdouts
6	Connect tracker (if used) · verify board truth
7	Host readiness · fence / waiver honesty
8	Assign issue · interpret terminal states · accept / merge

Any work day · four control surfaces

LANE

How much ceremony?

cv g lane – FAST / NORMAL / FULL with hard floors

NEXT

Where am I?

cv g next – evidence board + prompt + gate

GATE

Is this pass true?

cv g capture|intent|structure|...

TOKEN

Can I trust it?

One greppable line. Agents and owners read the same truth.

01

Start session

cv g setup healthy → cv g lane → cv g next --lane ...

02

During pass

Agent follows pass-prompt. You answer decisions only. Gate token green before moving on.

03

Build half

cv g ready → bind → loop. Never edit sealed evals to force green.

DONE IS PROVEN.

36 / 38

You are converged when the eval passes – not when you feel done.

DESIGN

Precision first

Cheap kills early. Soft intent poisons every later pass.

BARRIER

Your signature

Cross-family attack + human sign-off before machines build.

BUILD

Named states only

SETTLED is success. EXHAUSTED is a landing, not a win.

Chat is the UX. Disk + tokens are the truth.



END OF HANDBOOK

Done is proven.

Converge · Owner's handbook · complete edition v2

COMPILE INTENT INTO SHIPPED SOFTWARE.

CLI · CVG SKILLS · 12 PASSES · 0-8 IDENTITY · SETTLEMENT FOLD